

Basic Image Arithmetic and Point Operators

Prof. Jungong Han (jungong.han@sheffield.ac.uk)
Department of Computer Science
The University of Sheffield

- Which is/are true with respect to computer vision?
 - Deals with how to use computers to gain high-level understanding from digital images or videos
 - It seeks to automate tasks that the human visual system can do
 - It's always better than the human visual system
 - Is the same as image processing
- Sampling
 - Affects image resolution
 - Refers to the possible values that a reading may take
 - Is related to how many readings (in space) are made from a scene
 - Is related to how many sensing elements the camera sensor has

Outline

- Basic Image Statistics
- Operating with other Images
- Point Operators
- Histogram Processing



Images Are Matrices of Discrete Numbers

- The spatial domain of the images will, usually, be defined as:

$$\Omega = \{(x, y) : 0 \leq x < N_{\text{cols}} \wedge 0 \leq y < N_{\text{rows}}\} \subset \mathbb{Z}^2$$

- Because images are matrices, instead of (x, y) , in our programs we must use (y, x) (or (r, c)) to index intensities

$$i = I(r, c)$$

- Also remember that intensities will, usually, be integer

$$i \in [0, 2^{\text{\#bits}} - 1] \subset \mathbb{Z}$$



011	027	034	035	064	076	090	114	147	155	128	114	089	080	060	052
025	034	037	040	081	093	104	125	161	163	146	164	145	124	090	064
046	056	061	069	096	118	134	136	128	130	118	162	154	134	110	065
090	101	108	112	119	130	126	109	121	129	087	157	168	142	118	097
084	093	102	111	119	125	115	086	108	117	083	149	164	145	112	093
081	096	112	126	138	135	134	097	104	141	120	150	154	154	131	102
077	089	102	115	122	116	128	105	107	127	144	138	132	133	112	087
069	078	089	098	099	104	110	102	092	073	079	077	079	085	087	075

Average

$$\mu_I = \frac{1}{N_{\text{rows}} N_{\text{cols}}} \sum_{r=0}^{N_{\text{rows}}-1} \sum_{c=0}^{N_{\text{cols}}-1} I(r, c) = \frac{1}{|\Omega|} \sum_{(r,c) \in \Omega} I(r, c)$$

where $|\cdot|$ is the cardinality of a set

Variance (and Standard Deviation)

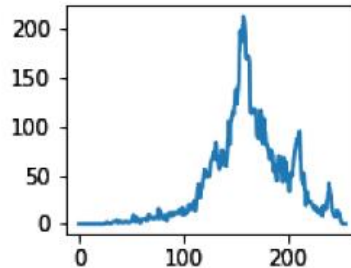
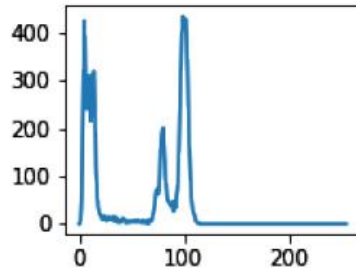
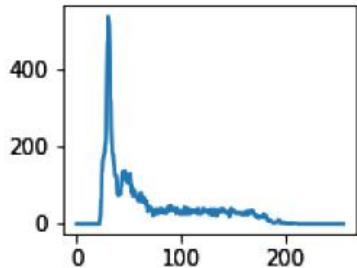
$$\sigma_I^2 = \frac{1}{|\Omega|} \sum_{(r,c) \in \Omega} [I(r, c) - \mu_I]^2$$

The standard deviation is just: $\sigma_I = \sqrt{\sigma_I^2}$

Image Histogram

Image histogram plots the number of pixels for each intensity value

$$H_I(i) = |\{(r, c) \in \Omega : I(r, c) = i\}|$$



```
Mat image = imread("picture.jpg", CV_LOAD_IMAGE_GRAYSCALE);

// allocate memory for no of pixels for each intensity value
int histogram[256];

// initialize all intensity values to 0
for(int i = 0; i < 255; i++)
{
    histogram[i] = 0;
}

// calculate the no of pixels for each intensity values
for(int y = 0; y < image.rows; y++)
    for(int x = 0; x < image.cols; x++)
        histogram[(int)image.at<uchar>(y,x)]++;
```

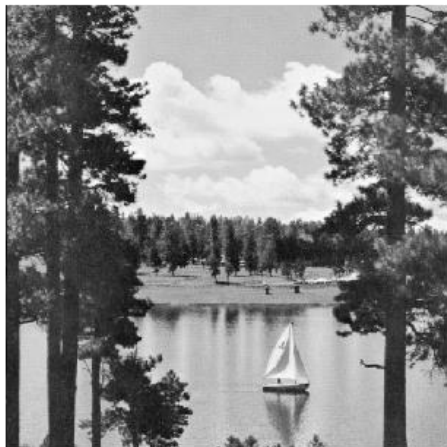
Review

- Images are matrices of (discrete) numbers
- The average of an image is one number representing the average intensity
- The variance and standard deviation indicates if intensities are generally close or far from the average (i.e. how compact is the histogram)
- The histogram tells how many pixels have a given intensity

Addition

$$I = \min(I_1 + I_2, \text{MAX})$$

I_1



I_2



$I = \min(I_1 + I_2, 255)$



Weighted Addition

$$I = \min(\alpha I_1 + \beta I_2, \text{MAX})$$

I_1



I_2



$I = \min(\alpha I_1 + \beta I_2, 255)$

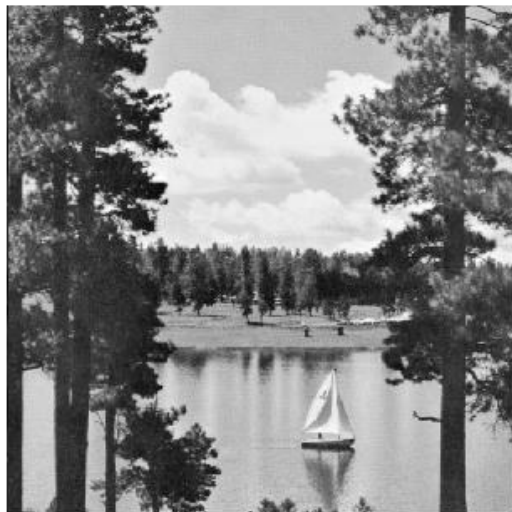


Alpha-blending: $I = \alpha I_1 + (1 - \alpha) I_2$

Subtraction

$$I = \max(I_1 - I_2, 0)$$

I_1



I_2



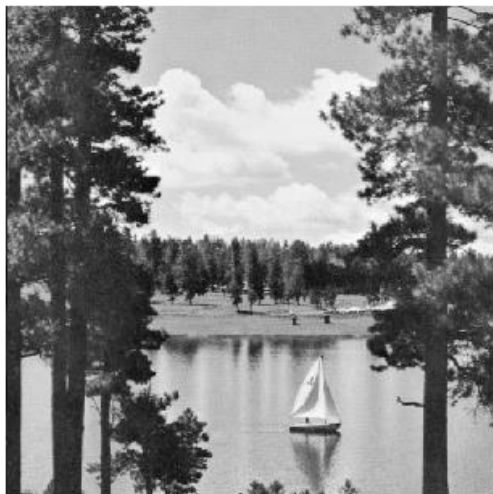
$I = \max(I_1 - I_2, 0)$



Absolute Difference

$$I = |I_1 - I_2|$$

I_1



I_2



$I = |I_1 - I_2|$



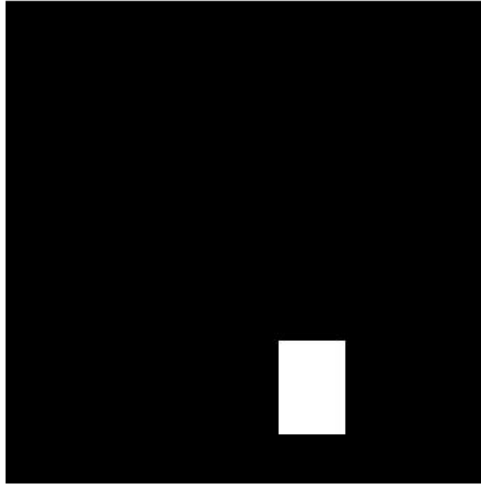
Multiplication

$$I = I_1 \times M$$

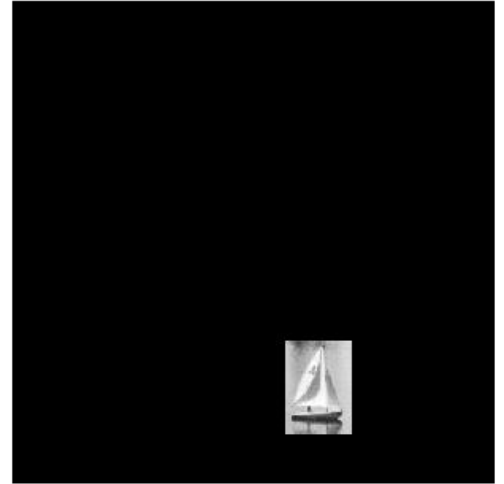
I_1



M



$I = I_1 \times M$



Point Operators

- AKA pixel transforms
- An operation h of the form:

$$I'(r, c) = h(I(r, c))$$

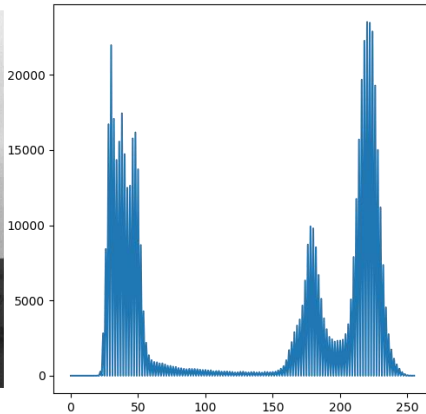
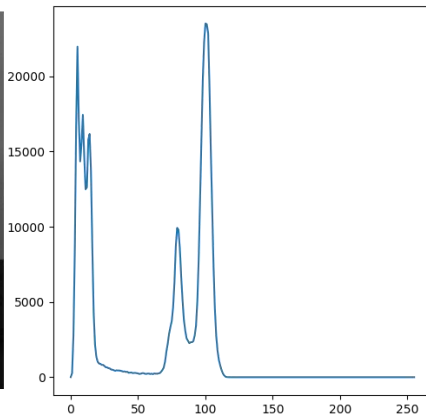
produces a new intensity solely based in the input intensity

- h frequently involves addition and/or multiplication

Brightness and Contrast

- This point operator has the following form: $i' = \alpha i + \beta$
 - i is the intensity
 - α controls the contrast (gain)
 - β controls the brightness (bias)
- Or, in matrix format: $I' = \alpha I + \beta$

Brightness and Contrast: $\alpha=2.0$; $\beta=20$

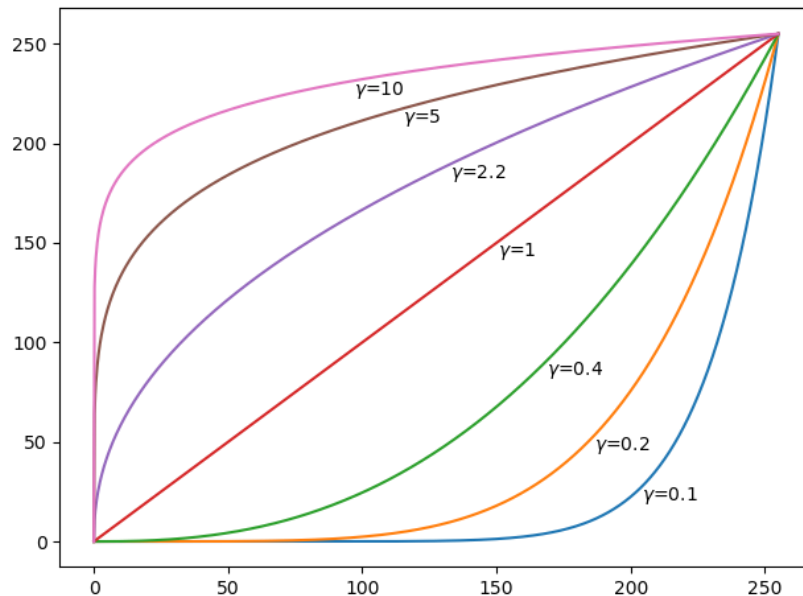


Gamma Correction

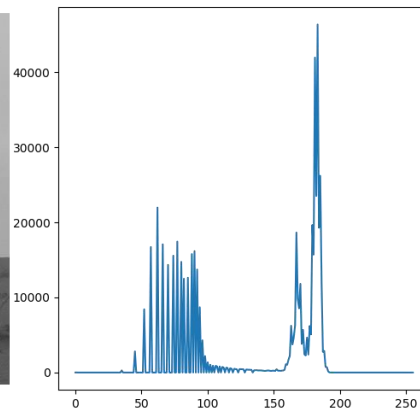
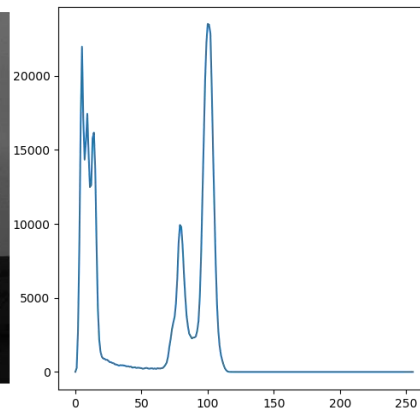
- AKA Power Operator, has the following form:

$$i' = \left(\frac{i}{255} \right)^{\frac{1}{\gamma}} \times 255$$

- $\gamma > 1$ darker regions become brighter
- $\gamma < 1$ darker regions become even more dark



Gamma Correction: $\gamma = 2.8$

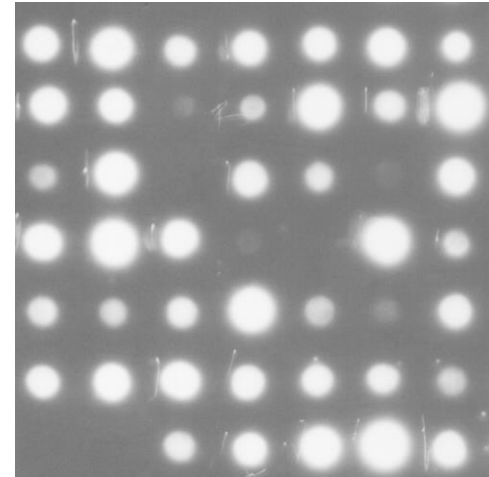
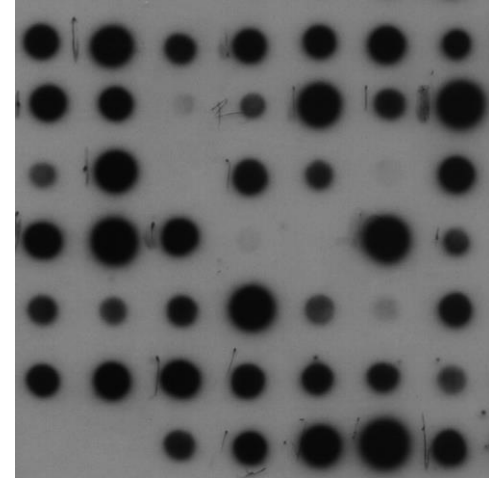


Intensity Inversion

- Equivalent to the photographic negative

$$i' = 255 - i$$

- Alternatively: bitwise not

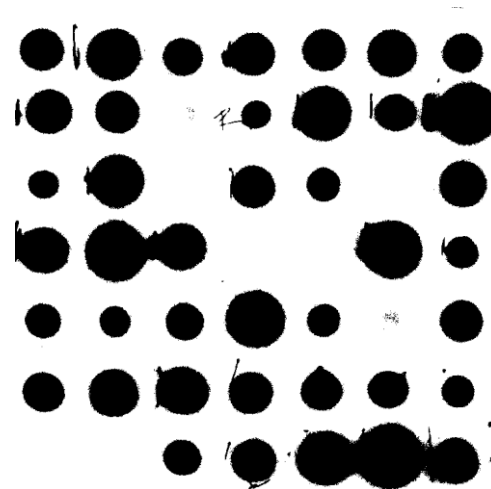
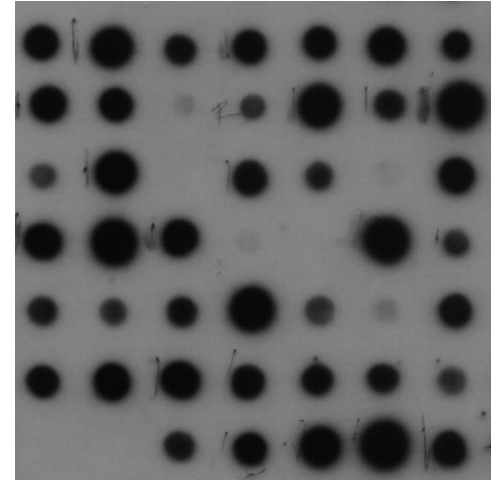


Thresholding

- Has the form:

$$i' = \begin{cases} 255, & \text{if } i \geq t \\ 0, & \text{otherwise} \end{cases}$$

- where t is the threshold level

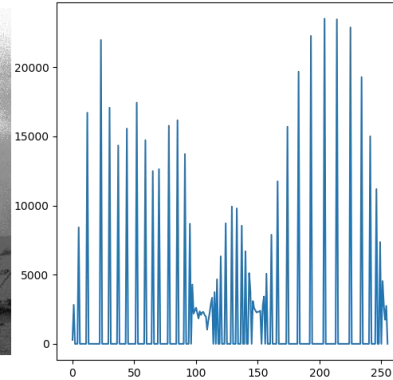
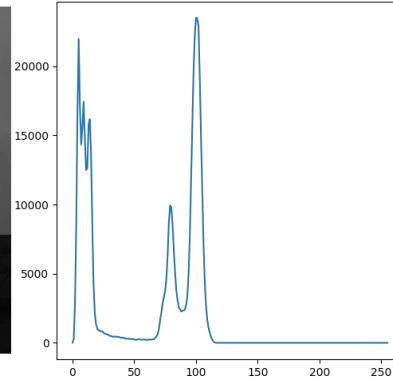


Histogram Equalization

- Tries to make all intensities equally represented (increase contrast)
- It uses the cumulative histogram H of the image
- Good for automatic image comparison
- May produce unnatural results

$$i' = \left\lfloor H(i) \times \frac{255}{m \times n} \right\rfloor$$

Histogram Equalization: Effect



Histogram Normalization

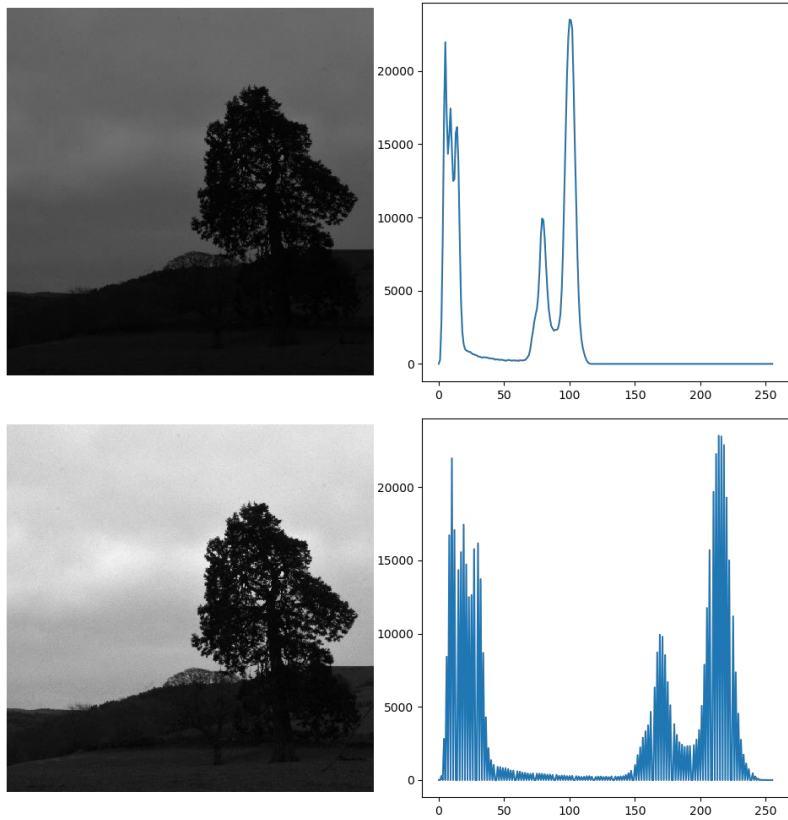
- AKA Histogram stretching
- Almost a point operator (requires 2 other intensities)

$$i' = \frac{i - i_{\min}}{i_{\max} - i_{\min}} \times 255$$

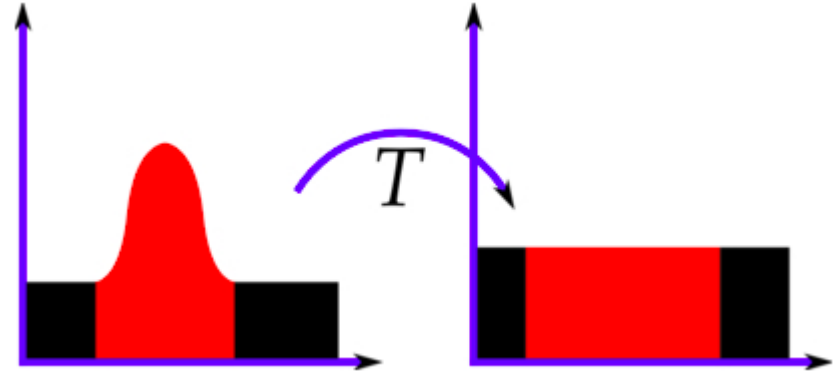
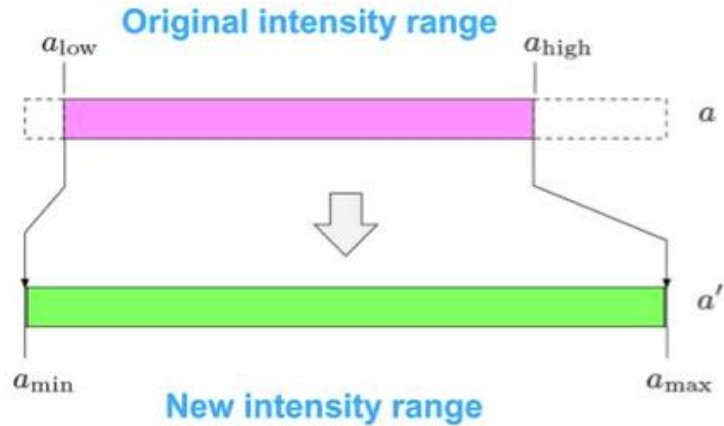
- where $i_{\min}$ and $i_{\max}$ are the smallest and largest intensity values present in the image, respectively

- A more general formula:
$$i' = \left(\frac{i - i_{\min}}{i_{\max} - i_{\min}} \right)^{\frac{1}{\gamma}} \times (o_{\max} - o_{\min}) + o_{\min}$$

Histogram Normalization: Effect



Histogram Normalization vs Equalization



Review

- A point operator uses the information of no more than 1 pixel
- Brightness is related to how strong we perceive the intensity of the pixels
- Contrast is related to how much dynamic range is available and used
- Thresholding consists in “selecting” pixels above an intensity level
- Histogram equalization changes the intensities in the image so that its histogram is as uniform as possible
- Histogram normalization “stretches” the histogram so that pixels will have a range of intensities from a minimum value to a maximum value